



Improving high school students' executive functions through digital game play



Bruce D. Homer ^{a, b, *}, Jan L. Plass ^b, Charles Raffaele ^a, Teresa M. Ober ^a,
Alisha Ali ^{b, c}

^a PhD Program in Educational Psychology, The Graduate Center, City University of New York, 365 5th Ave., New York, NY 10016, USA

^b CREATE Lab, New York University, 196 Mercer St., 8th Floor, New York, NY 10003, USA

^c Department of Applied Psychology, New York University, 196 Mercer St., 8th Floor, New York, NY 10003, USA

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ABSTRACT

Executive functions (EF), the skills required to plan, monitor and control cognitive processes, are linked to many important educational and developmental outcomes. The *Alien Game* is a digital game developed to train the EF subskill of *shifting*. High school students ($N = 82$; age range 14–18 years; average = 15.5 years) were asked to play the *Alien Game* for 20 min per week for 6 consecutive weeks. Two EF measures were administered before and after this intervention: the *Dimensional Change Card Sort* (DCCS) task (a measure of shifting) and the *Flanker* task (a measure of inhibition). Students had a significant pre- to posttest increase in DCCS, $t(81) = 4.29$, $p < 0.001$, $d = 0.54$, and Flanker, $t(77) = 2.93$, $p = 0.004$, $d = 0.22$. Controlling for pretest score, gains in shifting were significantly predicted by a measure of game performance in the *Alien Game*. These findings provide evidence that the *Alien Game* is having the intended effect of improving EF, and argue that video games can be effective tools for training cognitive skills when they are explicitly designed for this purpose and when a rigorous design approach is used.

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1. Introduction

Executive functions (EF) are the set of skills required to plan, monitor and control cognitive processes. EF, which develop throughout childhood into adolescence and early adulthood (Blakemore & Choudhury, 2006; Diamond, 2013), are related to a number of important outcomes, including externalizing behavioral problems (Young et al., 2009), social functioning (Miller & Hinshaw, 2010), and academic success (e.g., Blair & Razza, 2007; Yeniad, Malda, Mesman, van Ijzendoorn, & Pieper, 2013). Because of the importance of EF, there has been considerable research on the topic, and great interest in developing successful interventions for improving EF.

One of the challenges in studying EF is that a number of conceptually similar theories have been proposed to explain the cognitive systems involved in the planning, monitoring, and modifying of behaviors. In neuroscience, the ability to set goals, monitor outcomes, and adapt behaviors has often been referred to as *cognitive control* (e.g., Miller, 2000), which has been linked to the prefrontal cortex. For example, Norman and Shallice (1986) proposed a model of attentional control in which there is a *Supervisory Attentional System* (SAS) that is responsible for the planning, modifying and execution of actions.

* Corresponding author. Ph.D. Program in Educational Psychology, The Graduate Center, CUNY, 365 5th Ave, New York, NY 10016, USA.
E-mail address: Bhomer@gc.cuny.edu (B.D. Homer).

Similarly, [Baddeley's \(1986\)](#) model of working memory posits two slave systems, the *phonological loop*, for dealing with verbal information, and the *visuospatial sketchpad*, for dealing with visual and spatial data, in addition to a *central executive* that is responsible for monitoring and controlling the other systems.

A number of studies have attempted to identify the specific skills that are part of EF. Some of the first work in this area was done by [Luria \(1976\)](#), who observed behaviors of individuals who had suffered damage to their frontal lobes, particularly the prefrontal cortex, and noted that these individuals often demonstrated a lack of planning and greater impulsivity compared to how they were reported to have behaved prior to their injuries. This led [Luria \(1976\)](#) to propose that the frontal lobes were part of the functional unit of the brain responsible for regulating and controlling cognition and behavior. As part of this work, Luria developed a number of tasks to assess specific cognitive skills that are now associated with EF, including the go/no-go task, in which participants must respond to one type of stimulus but not another, and the non-verbal Stroop task, in which participants must respond to a picture in a counter-intuitive way, for example by saying “day” in response to a picture of the moon and “night” in response to a picture of the sun ([Hughes, 2005](#)). These are among the now dozens of tasks currently used to assess some component of EF (see [Chan, Shum, Touloupoulou, & Chen, 2008](#) for a review). There have been a number of efforts to standardize some of these EF measures, including the NIH Toolbox ([Weintraub et al., 2013](#)) which includes the two EF measures that were used in the current study: The *Flanker* task and the *Dimensional Change Card Sort* (DCCS) task.

The vast array of different skills and related measures have lead researchers to question whether EF should be considered a unified construct or a broader “umbrella” term for a diverse set of skills. Miyake and colleagues ([Miyake, Friedman, Emerson, Witzki, Howerter, & Wager, 2000](#)) used a latent variable approach to study unity or diversity of EF. They found three related, but separable components of EF: *shifting*, the ability to shift between tasks or mental sets; *inhibition*, the ability to suppress a prepotent response, and *updating*, the ability to monitor and manipulate elements in working memory. Each of these components was found to uniquely contribute to performance on complex cognitive tasks.

The [Miyake et al. \(2000\)](#) model of the three related, but separable components is arguably the most commonly used approach to understanding EF. There is evidence, however, that the nature of EF changes with children's cognitive development. [Ven, Kroesbergen, Boom, and Leseman \(2013\)](#) examined the three-factor model of EF in children at age 6 and again 18 months later. The researchers found that a two-factor model, consisting of an updating component and a combined inhibition/shifting component better fit their data than the three-factor model. Ven et al. suggest that the structure of EF may therefore change from childhood to adulthood, with inhibition and shifting separating later in development.

Regardless of the specific model of EF, there is considerable evidence of the importance of EF for educational outcomes. For example, [Blair and Razza \(2007\)](#) followed a group of 3- to 5-year-old children from low-income homes and gave them measures of various cognitive and academic skills at preschool, and again a year later, in kindergarten. They found that both inhibition and shifting in preschool uniquely predicted children's math and literacy skills in kindergarten. Similar findings were reported by [Bull, Espy, and Wiebe \(2008\)](#), who gave a battery of cognitive measures, including measures of EF, to children who were 4 years old, and then examined academic achievement, including math and reading measures, three years later, when the children were 7 years old. Bull et al. found that both working memory and EF at age 4 predicted math and reading performance at age 7. Although most of the research in this area has been done with younger children, the relation between EF and academic achievement continues into adolescence. [Best, Miller, and Naglieri \(2011\)](#) examined data from a large sample of children and adolescents (ages 5 through 17) and found that performance on EF tasks was significantly correlated with measures of reading and mathematics at all ages. Similarly, [Homer and Plass \(2014\)](#) found that children's performance on the Stroop task, a measure of inhibition, predicted learning outcome for high school chemistry students.

In their review of brain development during adolescence, [Blakemore and Choudhury \(2006\)](#) argue that the considerable physiological changes undergone by the brain during puberty and adolescence are responsible for dramatic changes in EF and social cognition. During adolescence, as with younger children, there continues to be a relation between EF and academic outcomes. For example, [Latzman, Elkovitch, Young, and Clark \(2010\)](#) examined the relation between different components of EF and academic outcomes in a group of adolescent boys. They found that flexibility predicted reading and science, updating predicted reading and social studies, and inhibition predicted mathematics and science. In a more recent study, [Rhodes, Booth, Palmer, Blythe, Delibegovich, and Wheate \(2016\)](#) examined the relation between adolescents' EF and chemistry learning and found that EF, particularly working memory, was associated with various chemistry learning outcomes.

A number of studies have also found that EF skills in adolescents and young adults are related to social skills, including theory of mind (ToM). ToM is the ability to attribute mental states to oneself and to others ([Apperly, 2011](#)). ToM initially develops in early childhood, and continues to develop in later childhood and adolescence, with increased automaticity and flexibility in reasoning about mental states ([Hayward, Homer, & Sprung, 2016](#)). A number of researchers have found EF to be related to development of ToM in childhood (e.g., [Carlson, Moses, & Claxton, 2004](#); [Hughes, 1998](#)) and in adolescence ([Dumontheil, Apperly, & Blakemore, 2010](#); [Vetter, Altgassen, Phillips, Mahy, & Kliegel, 2013](#)). Furthermore, impairments in EF, such as are found in children and adolescents with ADHD, are associated with impaired ToM, which is related to social functioning problems in adolescents ([Feigenbaum, 2017](#)).

Given the relation between EF and key developmental outcomes, there has been considerable effort to find interventions that improve EF skills. In their review, [Diamond and Lee \(2011\)](#) highlight a variety of activities that can improve children's EF, ranging from explicit classroom curricula to martial arts training. The authors highlight two key features of any successful intervention: repeated practice and progressively more challenging activities. Interestingly, computerized interventions, including training programs and digital games, have met with limited success even though repetition and “leveling up” are features of most digital programs and games.

Although recent reviews have found mixed results for the effectiveness of digital games as interventions for improving EF (e.g., Mayer, 2014; Powers, Brooks, Aldrich, Palladino, & Alfieri, 2013), there is still some evidence of their potential. One approach has been “brain training” games – essentially gamified versions of EF tasks. Nouchi et al. (2012) found significant improvements in EF in older adults who played the brain-training game, *Brain Age*, for 15 min a day, 5 days per week over a 4 week period as compared to a control group. However, other studies have not found significant effects. Owen et al. (2010) recruited a large online sample and found that participants who were trained with EF tasks got better only at those specific tasks – the effects did not transfer to other tasks. In general, meta-analyses suggest that the effects of brain-training interventions are small, and often are not significant (e.g., Lampit, Hallock, & Valenzuela, 2014).

More positive results have been found with off-the-shelf commercial digital games. For example, Green, Sugarman, Medford, Klobusicky, and Bavelier (2012) found that compared to a control group that played non-action games such as *Tetris*, a group of young adults who were randomly assigned to play intense action games, such as *Unreal Tournament* or *Call of Duty*, for 50 h over a 6–14 week period demonstrated significant improvements in the EF subskill of shifting. Similarly, Whitlock, McLaughlin, and Allaire (2012) found that a group of older adults who played *World of Warcraft* for 14 h over a two week period demonstrated significant improvements in inhibition. Other researchers have found similar positive effects of video game play (e.g., Basak, Boot, Voss, & Kramer, 2008; Glass, Maddox, & Love, 2013). On the other hand, a number of studies have reported negative effects. For example, Boot, Kramer, Simons, Fabiani, and Gratton (2008) found that young adults who played the action game *Medal of Honor* for 15 sessions, lasting 1–1.5 h each over the course of four to five weeks, did not demonstrate significant gains in EF. Furthermore, even studies that have found that playing commercial games can significantly increase some aspects of EF, have often failed to find effects for other aspects of EF, often without obvious explanations of why one EF skill would improve and another would not (e.g., Basak et al., 2008; Glass et al., 2013).

There are issues with both the brain training and commercial games approaches that may explain these findings. Brain training interventions are often just computerized or gamified version of standard EF tasks that incorporate some game elements rather than actual, fully-developed games. The gamified tasks often lack essential features for effective game-based interventions, including fully developed game mechanics, capabilities to adequately “level-up” in order to provide a consistent challenge to players, and a robust rewards system (e.g., one that goes beyond just “pass/fail”). These elements help well-designed games be engaging and motivating, which is essential for games to be effective tools for learning and cognitive skills development (Plass, Homer, & Kinzer, 2015).

On the other hand, interventions with commercial games may very well have all of the essential game elements required to be engaging and motivating for players, but they are unlikely to be effective if they are not designed with a specific learning objective in mind. Many commercial games may require some elements of EF in order for players to succeed, but because they are not designed explicitly to train EF skills, they will have limited success as an intervention.

We propose, therefore, that for a digital game to be an effective intervention for the development of EF, it must be both a fully developed game with the game elements that will allow it to be engaging and challenging for players, and also explicitly target a specific EF subskill. One recent example of this is *Neuroracer*, a custom-designed digital game involving driving a car in a 3-D environment while paying attention to a second task requiring responses to signs (Anguera et al., 2013). Older adults who played *Neuroracer* for 1 h per day, 3 times a week for 4 weeks demonstrated significant improvements in working memory and sustained attention. The likely reason for the effectiveness of *Neuroracer* is that its game mechanic, i.e., the essential, repeated game play activity, was designed specifically to train the targeted cognitive skill.

The current study examined the effectiveness of another custom-designed game, the *Alien Game*, that was intended to target the specific EF subskill of shifting in adolescent high school students. Research on an earlier version of the game found it to be effective with younger children (Leyrer et al., 2013). The current version was designed for older players, specifically adolescents, by providing increased challenge and more levels of play. It was hypothesized that after playing the *Alien Game* over a period of several weeks, participants would show significant improvements in the targeted skill of shifting. Pre- and posttest measures of another EF subskill, inhibition, were also given in order to examine the extent to which training would transfer from one EF subskill to another.

2. Material and methods

2.1. Participants

Participants were high school students from a public school in New York City. Inclusion criteria included being fluent in English, being between 14 and 18 years of age (average age = 15.5 years), and playing the *Alien Game* on average for a minimum of at least 5 min per session for the 6 sessions of the study (i.e., for at least 30 min overall). Of the initial sample of 91 participants, 3 students were excluded from the study for being older than 18, and 7 were excluded for failing to meet the minimum game-play time criterion of 30 min, resulting in a final sample of 82 (50 boys). The final sample size was greater than the minimum sample size we determined *a priori* based on research with an earlier version of the *Alien Game* (Leyrer et al., 2013). The students at the participating school were ethnically diverse, with 50% Hispanic, 27% African American, 18% Asian, and 5% White students, and with 88% of the students being eligible for free or reduced lunch.

2.2. Materials

To assess EF, two computerized measures of EF from the NIH Toolbox were used (Zelazo et al., 2013): the *Dimensional Change Card Sort* (DCCS) task, a measure of shifting, and the *Flanker* task, a measure of inhibition. In the DCCS, children are shown a series of images that vary along two dimensions – color and shape. Children are first asked to sort the cards into one of two boxes based on one dimension (e.g., color). For example, if shown a picture of a blue boat and asked to sort by color, they must press a key to place the image in the “blue” box. The rules then shift, and children must sort according to the other dimension (e.g., shape). For example, the blue boat must now be sorted into the “boat” box, regardless of color. After the rule switch, younger children will typically perseverate and initially sort by the old rules, and even older children will show delay after a rule switch. In the Flanker task, children must indicate whether an item that appears in the center of the screen is facing right or left. (For younger children, the items are fish – for older children, such as the ones in the current study, the items are arrows.) As potential distractors (i.e., “flankers”), similar stimuli appear on either side of the target stimuli. The flankers are either congruent (i.e., facing the same way as the target stimulus), or incongruent (i.e., facing the opposite direction). The measure of inhibition is how well children are able to ignore incongruent flankers. For both tasks, accuracy and reaction time are taken into account in scoring. Both tasks have been normed for early childhood through late adulthood, and have been shown to have good reliability and validity (Zelazo et al., 2013).

The intervention for the study was the *Alien Game*, a digital game developed through a collaboration between developmental psychologists and game designers. The game was designed to have similar cognitive requirements as the DCCS, specifically, requiring the player to consistently shift between increasingly complex sorting rules. In the game, hungry and/or thirsty aliens of different types need to be provided with food and drinks. The Aliens vary along two dimensions: color and number of eyes. Before each level, players are given specific rules as to which alien requires which food or drink item. For example, *blue aliens are hungry and need food, red aliens are thirsty and need drink*. After completing a level, the rule then reverses (i.e., the blue aliens are thirsty and the red aliens are hungry). As the game levels increase, the rules become increasingly more complex, including embedded rules (e.g., *the one-eyed red aliens and two-eyed blue aliens are thirsty*).

During the game, different aliens appear at the top of the screen and fly towards the bottom. Player must use a “food gun” located at the bottom of the screen to shoot each alien the correct item (according to the current level's rule) before it reaches the bottom of the screen. If successful, the alien becomes happy and the player receives points, with more points for a quicker response; otherwise, if the alien reaches the bottom of the screen or is given the wrong item it becomes sad/angry and points are lost. Players are given immediate feedback on their actions, are able to see their overall score, and can see a meter of overall “alien happiness,” (see Fig. 1). Other game elements include a musical score and sound effects, and difficulty that increases with higher levels. In the higher levels, rules become more complex, involving more shifting (e.g., the appearance of a “radiation storm” will temporarily reverse the rules for a level), and quicker reaction times are required because the aliens fall at a faster rate.

An iterative design approach was used to develop the *Alien Game*. This involved playtesting early versions of the game with small groups of adolescents. During playtesting sessions, players would give feedback using think-aloud protocols while playing, and then semi-structure interviews were conducted afterwards to ask about play experience and any difficulties. In addition, gameplay was recorded using screen-capture software. The think-aloud protocols and interviews were summarized and triangulated with field notes and results of video analyses of gameplay to identify any usability problems encountered by players. Based on the outcomes of the playtesting sessions, the *Alien Game* was modified, and then the new version would undergo another round of playtesting. This continued until the game developers and researchers felt satisfied that the game was enjoyable and had no usability issues for the players. This final version was used for the current study.



Fig. 1. Screen Shots from the Alien Game: 1a) Instructions for Players (left panel), 1b) Points Earned for Alien Fed Correctly (right panel).

2.3. Procedure

The study took place in the students' classroom, once a week over an 8 week period. All measures and the intervention were presented via a computer, with each student working on his or her own computer. On the first week, demographic questionnaires and the computerized DCCS and Flanker tasks were administered. The *Alien Game* was then introduced to the students, who created accounts in the experimenters' online testing system with usernames that the students chose themselves. They then played through the first, tutorial level to learn the basics of the game. For the next 6 weeks, students were asked to sign in and play for 20 min per week. Each week, the next two levels of the game were unlocked. Students' performance in the *Alien Game* was captured via a log file. Variables in the log file included: *time played*, *levels completed*, *reaction time* (i.e., the time between when an alien first appeared and when it was given food or drink), as well as *aliens saved* and *unsuccessful attempts* for each level and overall. At the end of each week, a "Leaderboard" email was sent to all of the students indicating the total number of aliens saved and total play time for all students combined, as well as the usernames of the students who had played the most minutes, saved the most aliens, and had the quickest average reaction times. On the final week, the DCCS and Flanker tasks were administered again, and students were given a posttest questionnaire asking about their experiences with the game.

3. Results

For analyses, students were grouped according to their age in years (i.e., 14, 15, 16, 17, or 18). The students played the *Alien Game* for an average of 81 min, "saving" an average of 2449 aliens (i.e., there was an average of 2449 successful trials). No statistically significant gender differences were found on any of the measures except for an item on the demographic questionnaire asking how much time per week students spent playing video games: girls reported an average of 9 h per week of video game play, while boys reported an average of 36 h per week, $t(78) = 5.03$, $p < 0.001$.

Age-group means for each measure are reported in Table 1. To determine if students' scores on the DCCS and Flanker tasks changed significantly from pre-to posttest, paired t-tests were conducted. As hypothesized, students' scores significantly improved on the DCCS task, $t(81) = 4.29$, $p < 0.001$, $d = 0.54$. Students' scores also increased significantly on the Flanker task pre-to posttest, $t(77) = 2.93$, $p = 0.004$, $d = 0.22$.

A *game performance* score was calculated for each student, which was the total number of aliens saved divided by total time playing. Therefore, the higher the game performance score, the more aliens were saved per minute of play by a student. This metric of game performance reflects both effort and ability of the students.

Correlations between the variables are reported in Table 2. Interestingly, pretest measures of EF were not significantly related to one another, however, posttest measures of EF (i.e., posttest DCCS and Flanker scores) were significantly correlated with one another. Age was significantly negatively correlated with playtime, indicating that the older students played for less time than the younger students. Also, the total number of hours students reported typically playing video games during a week (*weekly video game play*), was not significantly related to any of the EF measures. Finally, the game performance measure was not significantly correlated with the pretest measures of EF (i.e., Pre-DCCS and Pre-Flanker), indicating that this was not just a "stealth assessment" (Shute, 2011) of EF. However, game performance was significantly correlated with posttest measures of shifting (i.e., Post-DCCS), indicating a relation between students' success in the game and their subsequent EF scores.

Next, a hierarchical linear regression was computed in order to examine which factors predict posttest scores on the DCCS. With DCCS posttest scores as the dependent variable, pretest DCCS score was entered in the first step, and both total hours of *Alien Game* played and *Alien Game* performance measure entered as predictors in the second step. Also included in the second step was the total number of hours students reported typically playing video games per week, because the correlation of this variable to posttest DCCS had approached significance ($p = 0.09$). The results of the regression for the DCCS posttest score are reported in Table 3. Not surprisingly, pretest score accounts for a significant proportion of the variance in posttest DCCS score. There was a significant gain in R^2 in the second step of the model, with in-game performance accounting for a significant proportion of the variance in posttest scores, after accounting for pretest score.

A similar regression was conducted for Flanker posttest. However, in order to determine if there was transfer from training of one EF subskill to another, in other words, if gains in Flanker score (i.e., inhibition) could be attributed to gains in DCCS score (i.e., shifting), DCCS pretest score was entered in step one, and DCCS posttest score was entered in step two of the regression.

Table 1
Means (SD) on EF Measures and Playtime by Age Group.

Age (years)	Pre-DCCS	Post-DCCS	Pre-Flanker	Post-Flanker	Playtime (minutes)
14 ($n = 28$)	7.75 (2.04)	8.16 (1.37)	9.44 (1.23)	9.62 (0.65)	96 (29)
15 ($n = 20$)	7.15 (2.36)	8.65 (1.32)	9.65 (0.62)	9.88 (0.18)	97 (34)
16 ($n = 16$)	6.74 (2.61)	8.34 (1.11)	9.55 (0.67)	9.76 (0.44)	64 (29)
17 ($n = 11$)	6.51 (2.66)	7.64 (1.97)	8.35 (2.28)	8.91 (1.82)	52 (13)
18 ($n = 7$)	6.66 (2.62)	7.87 (2.72)	8.71 (2.25)	9.07 (2.35)	65 (26)
TOTAL ($N = 82$)	7.15 (2.36)	8.22 (1.55)	9.29 (1.39)	9.56 (1.07)	81 (33)

Table 2
Correlations Between Variables.

	1.	2.	3.	4.	5.	6.	7.
1. Pre-DCCS	—						
2. Post-DCCS	0.387*	—					
3. Pre-Flanker	0.199	0.469*	—				
4. Post-Flanker	0.159	0.438*	0.763*	—			
5. Playtime (minutes)	0.010	0.024	0.002	0.059	—		
6. Game Performance	-0.176	0.223*	0.013	-0.083	-0.049	—	
7. Age (y)	-0.191	-0.101	-0.233*	-0.211	-0.453**	0.171	—
8. Weekly Video Game Play (hours)	0.071	0.190	-0.065	0.008	-0.043	0.198	-0.133

Note: * $p < 0.05$; ** $p < 0.01$.

Table 3
Summary of Hierarchical Regression Analysis for Variables Predicting DCCS Posttest Scores (N = 82).

	Model 1			Model 2		
	B	SE B	β	B	SE B	β
Pre- DCCS	0.261	0.069	0.392***	0.290	0.067	0.435***
Total Playtime				0.134	0.280	0.047
Game Performance				1.89	0.692	0.281**
Weekly Video Game Play				0.006	0.006	0.106
R ²	0.153			0.254		
F for ΔR^2	14.31***			3.43*		

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Because total hours of reported weekly video game play was not correlated with either pre- or post-Flanker scores, game play time was not included in the regression. The results of the regression equation for the Flanker posttest score are reported in Table 4. Not surprisingly, pretest score accounted for a significant proportion of the posttest Flanker score in step one of the equation. There was a significant gain in R^2 in the second step of the model, with game performance accounting for a significant proportion of the variance in posttest scores, although this relation was negative, suggesting that students who had lower levels of in-game performance (i.e., who “saved” aliens at a slower rate) had greater transfer of EF subskills. Posttest DCCS score also accounted for a significant proportion of the variance in posttest Flanker score, after accounting for pretest Flanker and DCCS scores, indicating that gains in inhibition (Flanker) were related to gains in shifting (DCCS).

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

4. Discussion

The current study provides evidence that digital games that are both fully developed games and that have been custom built to target specific cognitive skills can be effective training tools for cognitive skills: Students' playing of the *Alien Game* had the intended cognitive effect of improving the EF subskill of shifting, as indicated by their significantly improved performance on the DCCS. The increase on the DCCS from pretest to posttest represents an effect size of $d = 0.54$, which is comparable to what is typically found for EF intervention studies, which range from $d = 0.48$ to $d = 0.61$ (Powers et al., 2013). Although these findings are limited by lack of a control group, the average gain on the DCCS, of 1.07 on the 10-point scale, is far greater than the 0.55 gain between ages of 14–18 years that is reported in the NIH Toolbox normed data (Slotkin et al., 2012), strongly suggesting that the gains in shifting cannot be attributed to developmental change occurring over the 8 weeks of the study. Furthermore, the fact that gains in the subskill of shifting were predicted by students' in-game performance provides evidence that changes in EF are due to the *Alien Game* intervention: The better that students did in the *Alien Game*, the greater were their gains in the EF subskill of shifting.

Table 4
Summary of Hierarchical Regression Analysis for Variables Predicting Flanker Posttest (N = 77).

	Model 1			Model 2		
	B	SE B	β	B	SE B	β
Pre-Flanker	0.564	0.057	0.756***	0.512	0.061	0.686***
Pre-DCCS	-0.016	0.035	0.036	-0.032	0.038	-0.071
Post-DCCS				0.135	0.062	0.203*
Total Playtime				0.100	0.135	0.053
Game Performance				-0.872	0.350	-0.195*
R ²	0.584			0.631		
F for ΔR^2	52.66***			3.05*		

The results from the current study have significant theoretical and practical implications. On the theoretical side, our results shed some light on the inconsistent findings from previous research on game-based interventions to improve EF. The *Alien Game* is a fully developed game, not just a *gamified* test (i.e., the same task as the test, but with rewards such as points or stars added). Well-designed games maximize player engagement and challenge, whereas *gamified* tests are not only unlikely to be interesting or challenging enough to maintain player attention and sustained efforts, their game mechanic is often not optimized to train the targeted skill since the test was designed to measure, not to train. Similarly, commercial games are typically not designed to train specific cognitive skills. However, with the *Alien game*, the entire game, including the game mechanics, reward system, levels, etc., was designed to target the specific EF subskill of shifting. Therefore we believe that by targeting a specific cognitive skill and by being a fully developed game, the *Alien Game* was a successful intervention.

Results from the current study also indicate that gains in shifting can transfer to the EF subskill of inhibition. Students' scores on the Flanker task improved significantly, and these gains were related to posttest scores in DCCS, after accounting for pretest scores on the Flanker and DCCS. Research on transfer of EF training has been mixed, with some studies finding evidence of transfer (e.g., [Karchach & Kray, 2009](#); [Thorell, Lindqvist, Bergman, Bohlin, & Klingberg, 2009](#)), and others finding little to no transfer from one EF subskill to another (e.g., [Baniqued et al., 2014](#); [Melby-Lervåg, Redick, & Hulme, 2016](#)). The current findings support the claim that there can be transfer from one EF subskill to another, but further research is needed to determine the factors that enable this near-transfer. One suggestion has been that a more complex training task is required for transfer to occur ([Mayer & Wittrock, 2006](#)), which may explain why a fully developed game is more likely to result in transfer than gamified tasks. A related question is to what extent training of EF skills will transfer to real-world applications, such as improved academic performance. There is some initial evidence of this far-transfer. In their review, [Titz and Karchach \(2014\)](#) conclude that there is evidence of academic benefits as a result of working memory and EF training, particularly for language and reading, with weaker support for benefits for math. Further research is needed to examine factors that support this far-transfer to real world applications.

The fact that students' performance on the *Alien Game* was negatively related to posttest Flanker scores (after accounting for pretest scores) is an unexpected finding. A possible explanation for this may have to do with the relation between students' in-game performance and inhibition. Students with low levels of inhibition may have been more distracted while playing the *Alien Game* and would therefore have had lower scores on the game performance measure. However, the students who started with lower levels of inhibition had the potential to make the greatest gains from pre- to posttest Flanker scores (i.e., they start with low pre-Flanker scores and therefore have more room for improvement). Therefore, the students who performed less well on the *Alien Game* (which had the goal of training switching) could be gaining the most pre- to posttest on the Flanker task (i.e., inhibition), resulting in the negative relation between game performance and posttest Flanker score. Further research is required to examine this possibility.

It is also worth noting that there was a significant, and rather large, gender difference in the amount of reported weekly video game play time. Previous studies have found that the gender difference in amount of video game play is becoming less pronounced for younger children (e.g., [Homer, Hayward, Frye, & Plass, 2012](#)), but it seems to persist for older children. It is notable, however, that girls in the current study are still spending a fair amount of time playing video games, reporting an average of 9 h of play per week. The lack of relation between overall video game play and any EF measure is also notable. This further supports the argument that it is not general video game play per se, but playing of specific games that can lead to improvement of EF and other cognitive skills.

Additional research is needed to determine if the results from the current study can be replicated for other EF subskills. Considering [Miyake et al.'s \(2000\)](#) three EF subskills, the current study finds support for use of digital games to train shifting. Other studies have found similar results with older participants using commercial action video games that require considerable amounts of shifting of attention during game play (e.g., [Green et al., 2012](#)). Similarly, [Whitlock et al. \(2012\)](#) found that playing 14 h of the commercial game *World of Warcraft* significantly improved older adults' performance on a Stroop task, which is a measure of attentional control and inhibition. Finally, positive effects of training have also been found for working memory. For example, a number of studies have found significant effects of *Cogmed* (Pearson Education, 2011), which was developed to train working memory (e.g., [Beck et al., 2010](#); [Klingberg, 2010](#)—but see also [Shipstead, Redick, & Engle, 2012](#) who call for more rigorous testing of CogMed). Additionally, [Anguera et al. \(2013\)](#) found in their research that older adults who played *NeuroRacer* also demonstrated improvements in working memory.

In their meta-analysis, [Powers et al. \(2013\)](#) point out that there has been very little work with children and adolescents on the use of video games to improve cognitive skills, including EF. For their analysis, Powers et al. had to combine studies with participants ages 7–17 into one group, which covers a broad developmental spectrum. They found that, although there were significant overall effects for all age groups, the effects for this youngest group (7–17 years) was significantly smaller, meaning that it is more difficult to find significant effects for children and adolescents. The current study is one of the few to explicitly target the use of video games to support the development of EF in adolescents.

On the practical side, the current study suggests that digital games can be an effective way to train cognitive skills, such as EF. The findings point to the benefit of using games that target specific cognitive skills. This is most easily done through the use of custom built games, as in the current study, but could also be done through the strategic use of commercial games. The findings also indicate the importance of building games that are genuine games, not just gamified cognitive tasks. By using fully developed, custom designed games, players are more likely to be cognitively and emotionally engaged in playing the game, which is an essential requirement for any game for learning or skill training.

The current findings also have implications for the use of games as assessment tools (Bellotti, Kapralos, Lee, Moreno-Ger, & Berta, 2013; Plass et al., 2013; Shute & Ke, 2012). As Gee (2013) points out, games can be an exemplary method of assessment: To succeed in a game requires application of skills and knowledge in ways that can be directly analogous to real life. Shute (2011; Shute & Ke, 2012; Shute & Sun, *in press*) draws upon evidence centered design (Mislevy & Haertel, 2006) to discuss the use of embedded assessment within games. Shute and her colleagues refer to this approach as “stealth” assessment because the assessment is not obvious to the learner, who is just actively playing a game. With stealth assessment, Bayesian networks are used to determine which specific in-game performance metrics relate to a particular measure of interest (Shute & Kim, 2014, pp. 311–321). For example, Shute, Ventura, and Kim (2013) used stealth assessment to examine how the ways that students problem solve in a commercial video game can be used to evaluate their conceptual understanding of physics.

In the current study, although the Alien Game performance measure predicts posttest scores of EF, it is not correlated with pretest EF scores, indicating that it is not useful as an assessment tool. It is possible that with further analyses, some in-game metrics may be found that predict pre-test EF scores (e.g., error rates and/or reaction time immediately following a switch in rules), but it is also possible that a game of this type, designed solely for the purpose of training EF skills, is not ideal for use as an assessment tool.

5. Conclusion

The *Alien Game* was a custom-built video game intended to support the development of the EF subskill of shifting in adolescent high school students. The game was developed with two goals, to ensure that it targeted the specific cognitive skill that was intended to be trained, and that it was fun for adolescents to play. After a six-weeks of game play, the students showed significant gains in the target EF subskill of shifting, and some transfer to the EF subskill of inhibition. These results indicate that video games can be effective tools for training of cognitive skills when they are explicitly designed for this purpose (rather than simply gamifying a cognitive skills measure), and when a rigorous design approach is used that involves playtesting and other methods to optimize the design of the games to train cognitive skills.

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